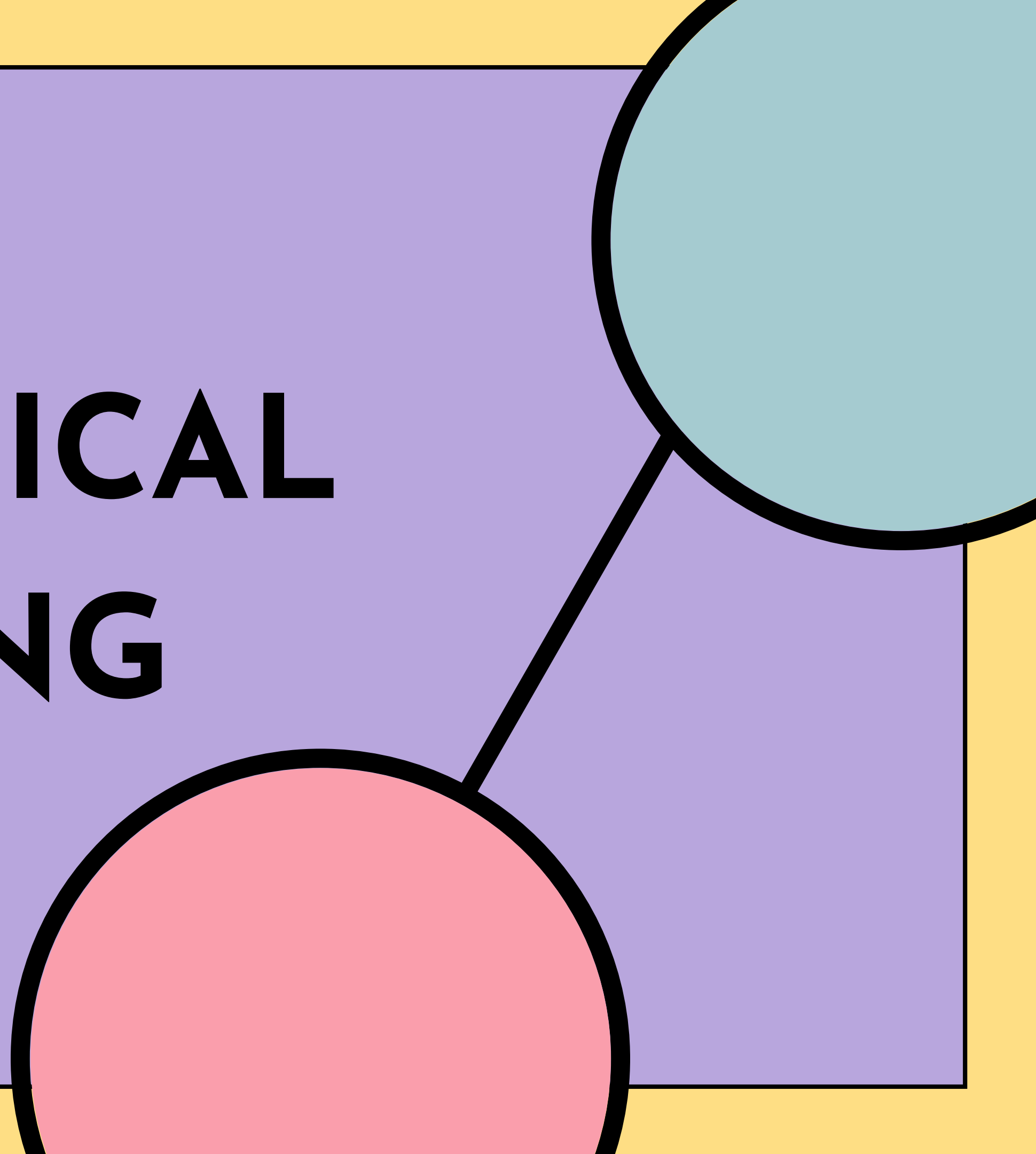


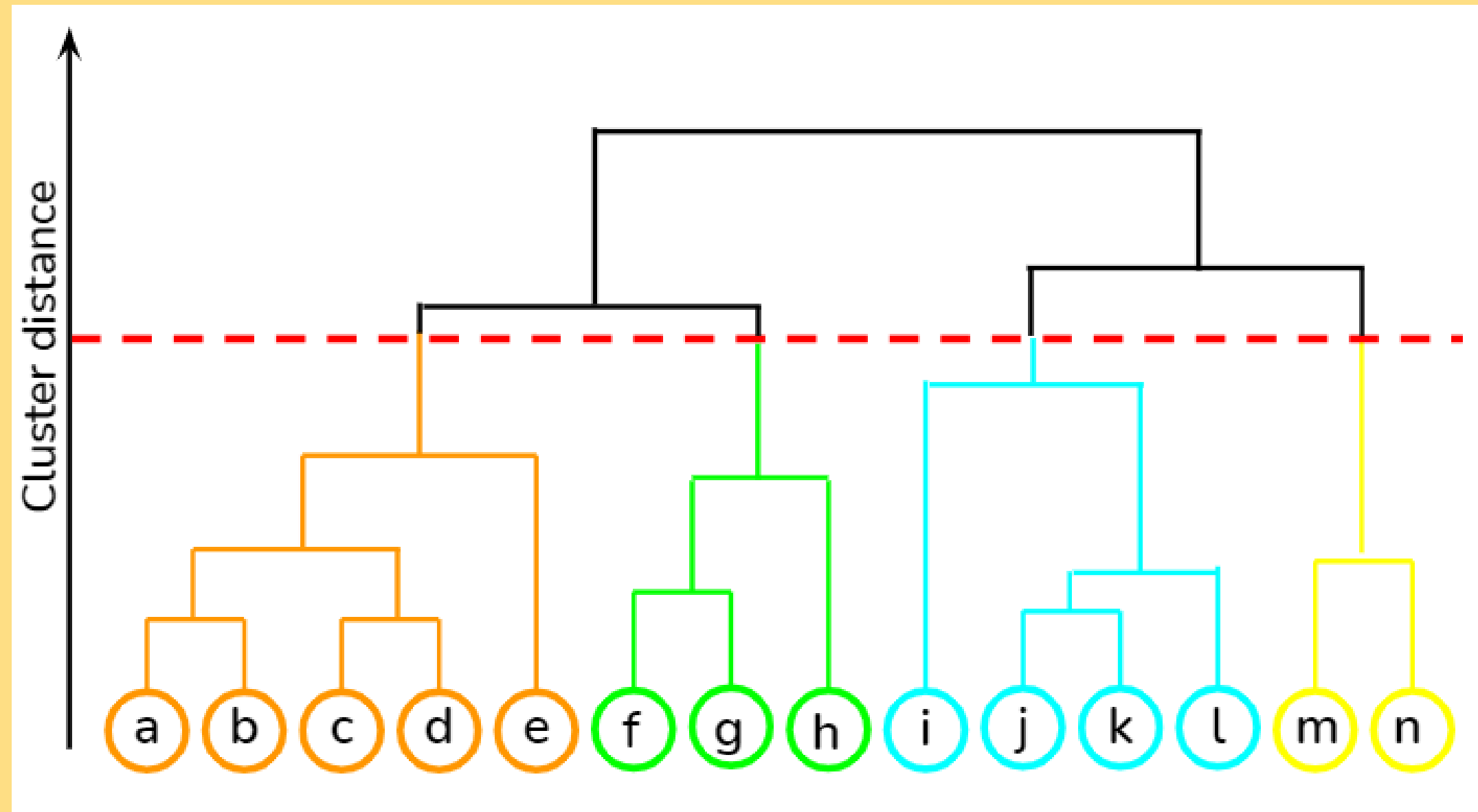
# **HIERARCHICAL CLUSTERING**



# What is Hierarchical Clustering?

- It is a clustering method that creates a hierarchical tree of objects to be clustered hierarchically.
- The tree represents the relationships between objects and shows how objects are clustered at different levels.

# What is Hierarchical Clustering?



**Dendrogram**

**How does Hierarchical  
Clustering work?**

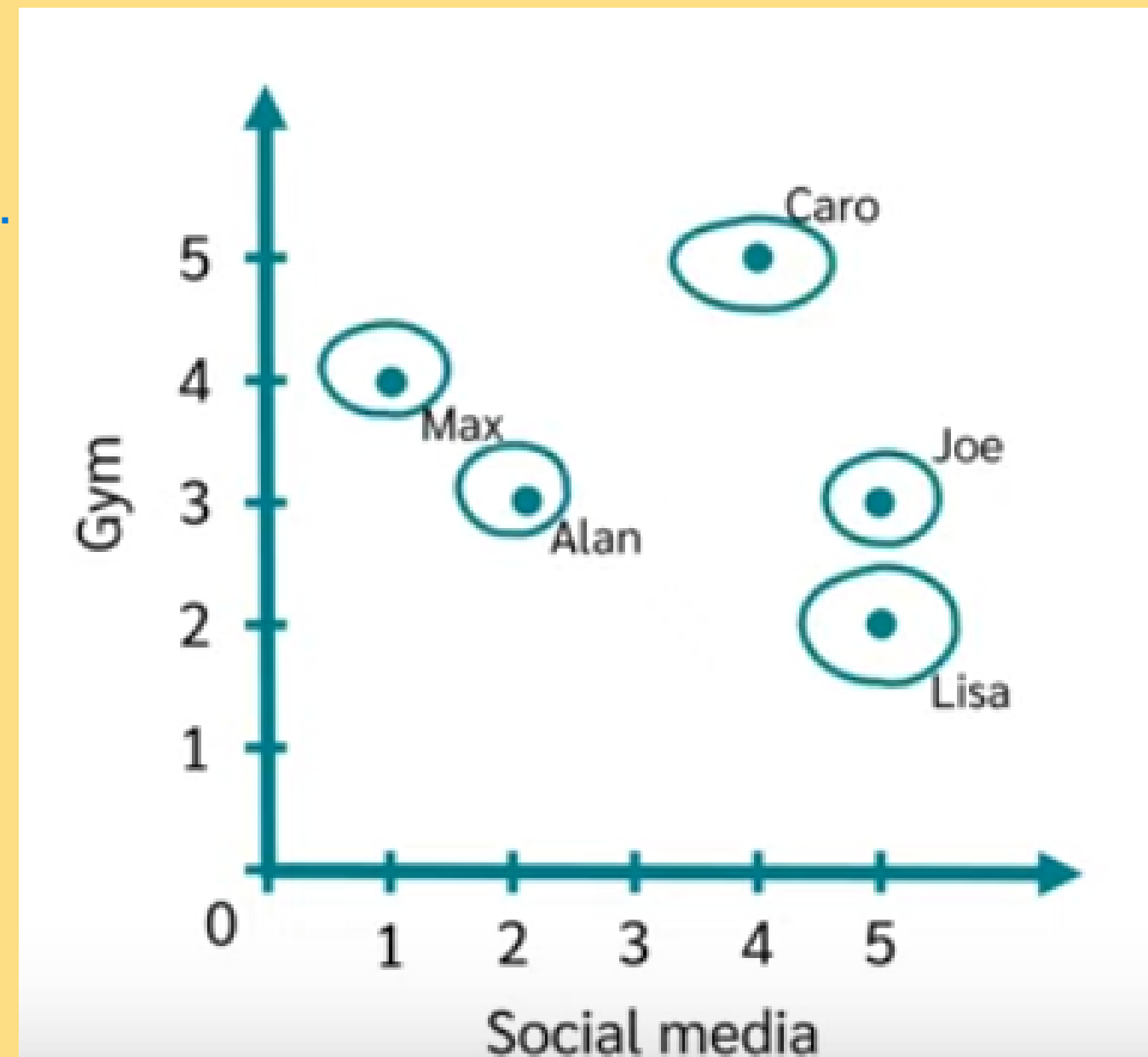
# 1- Assign a cluster to each individual point (number of clusters = number of points)

Agglomerative Clustering?

Bottom-up hierarchical clustering.

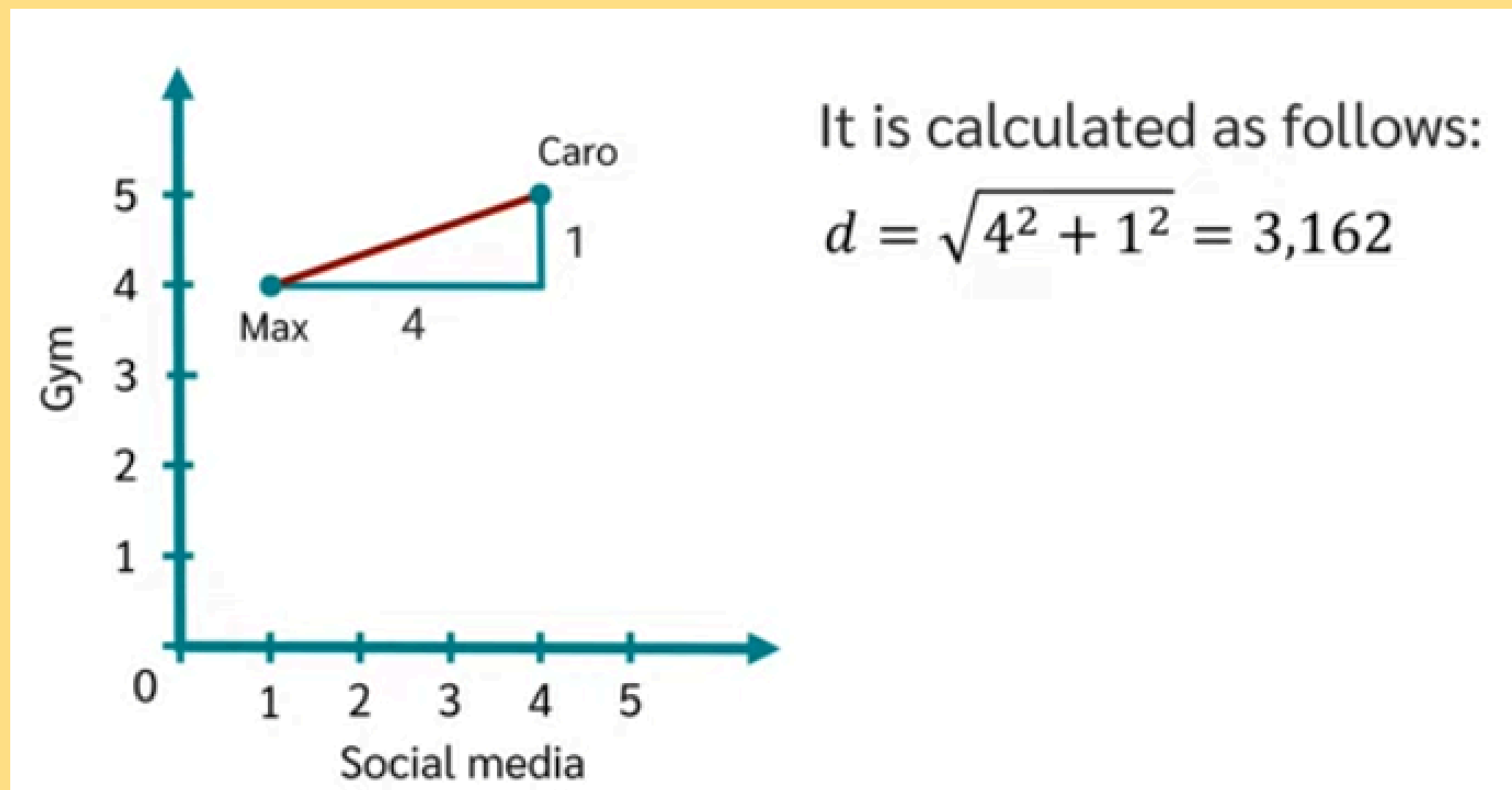
Divisive Clustering?

Top-down hierarchical clustering.



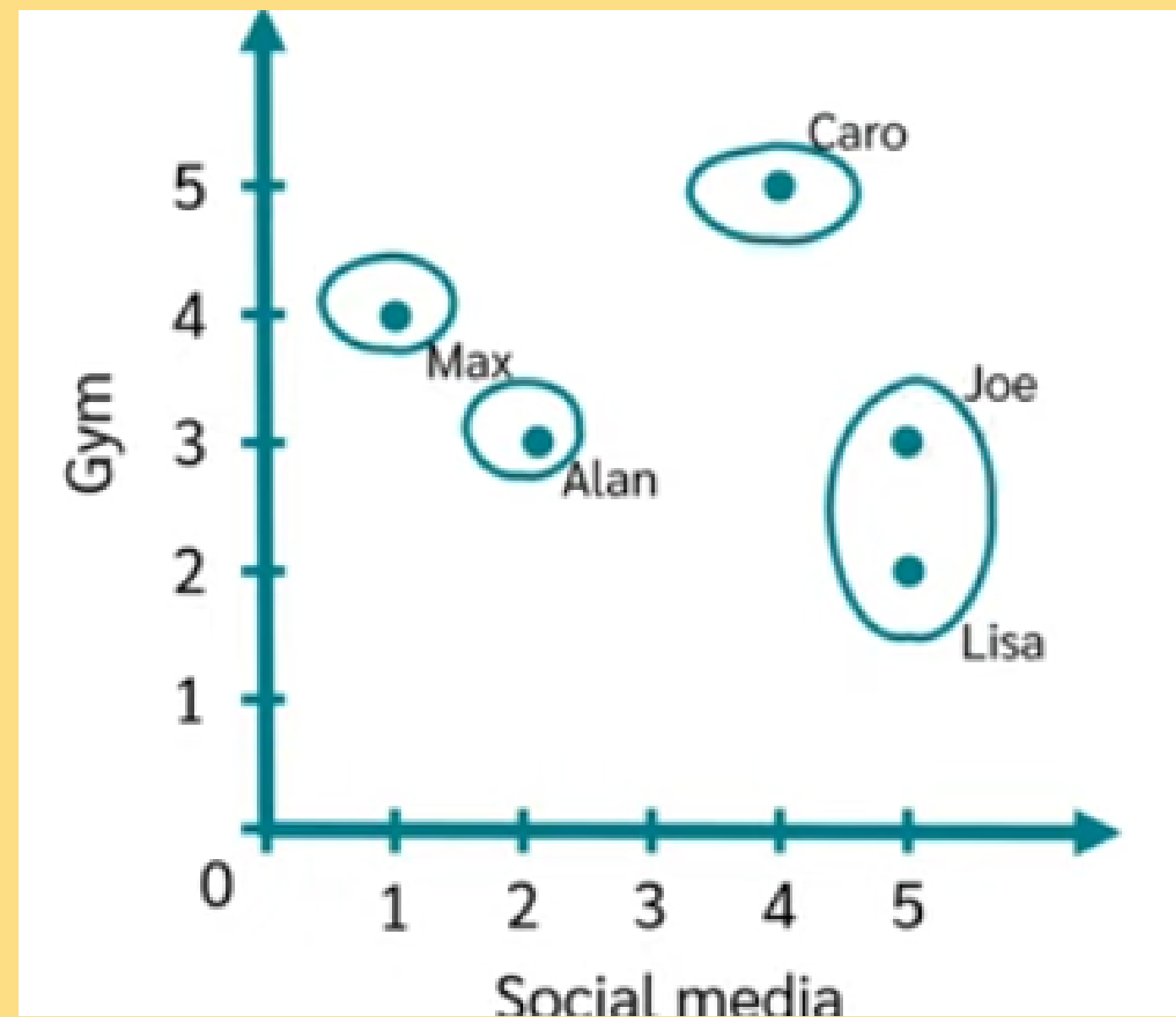
## 2- Calculate Pairwise distances

- Compute the distance between every pair of clusters using a distance metric .



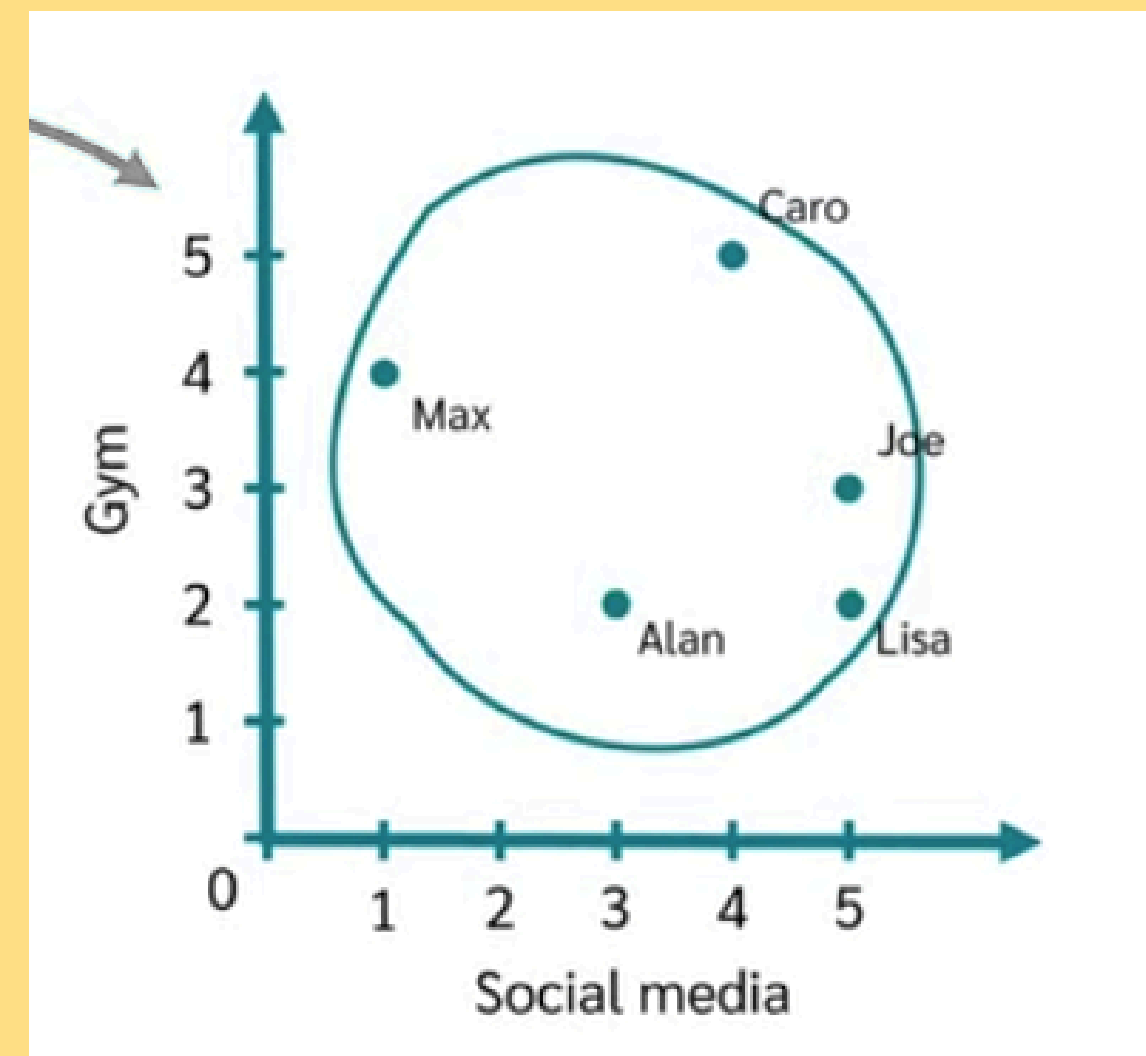
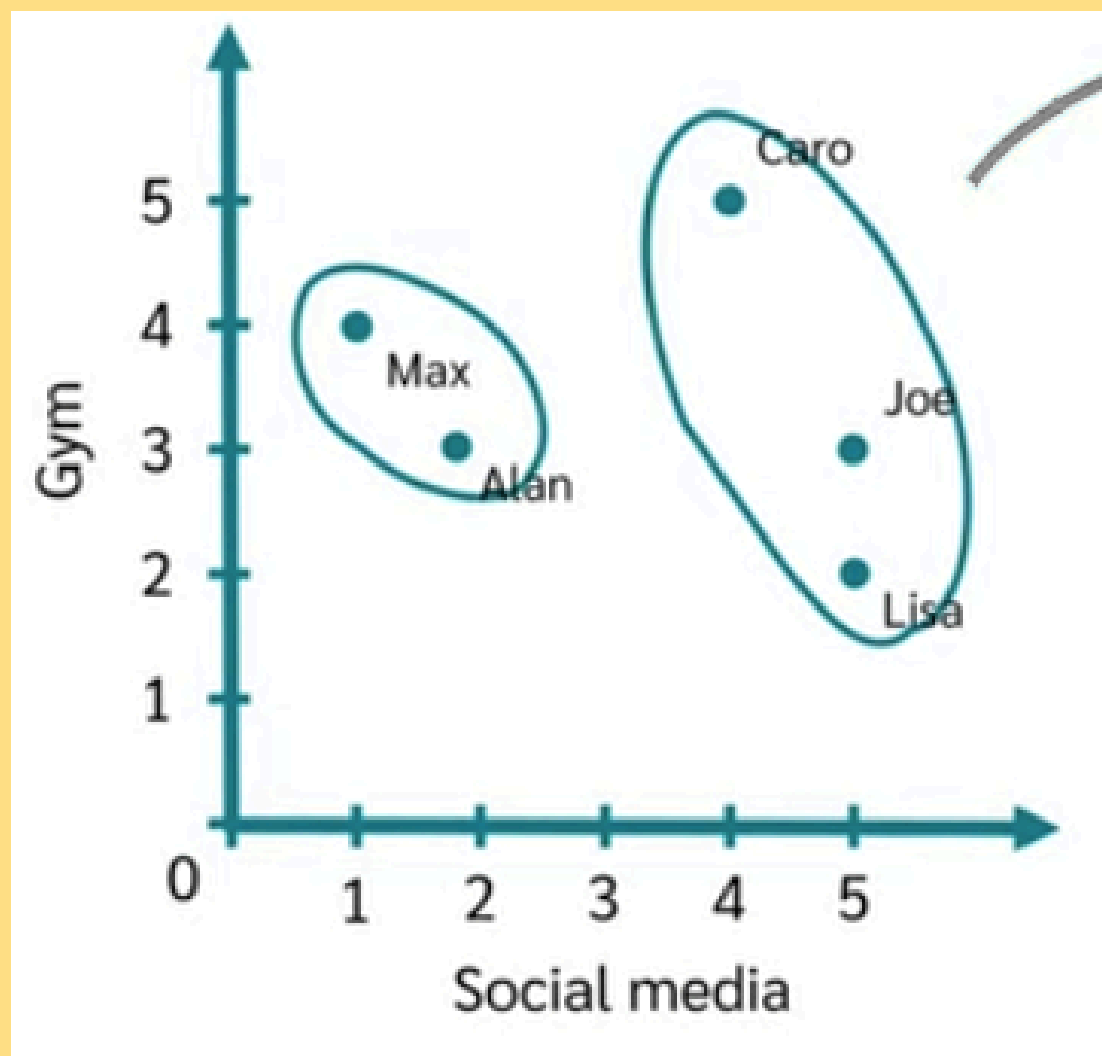
### 3- Link clusters

- Identify the two clusters that are closest to each other based on the distance metric, and merge these two clusters into a single cluster .



## 4- Repeat the process

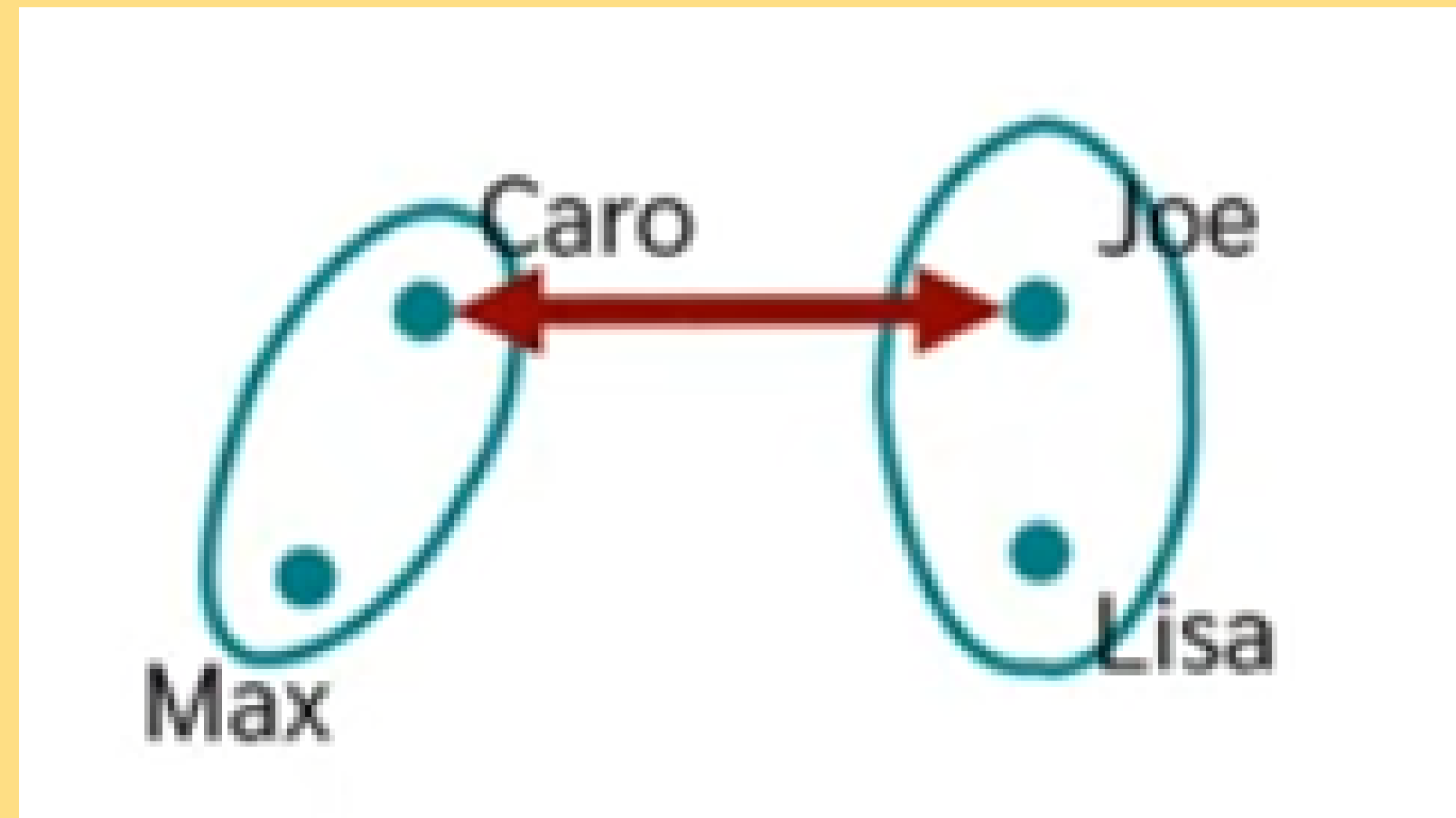
- Continue merging the closest clusters and updating the distance metric until all data points are merged into a **single cluster**, or until the desired number of clusters is reached.



**How to link cluster with  
more than one point?**

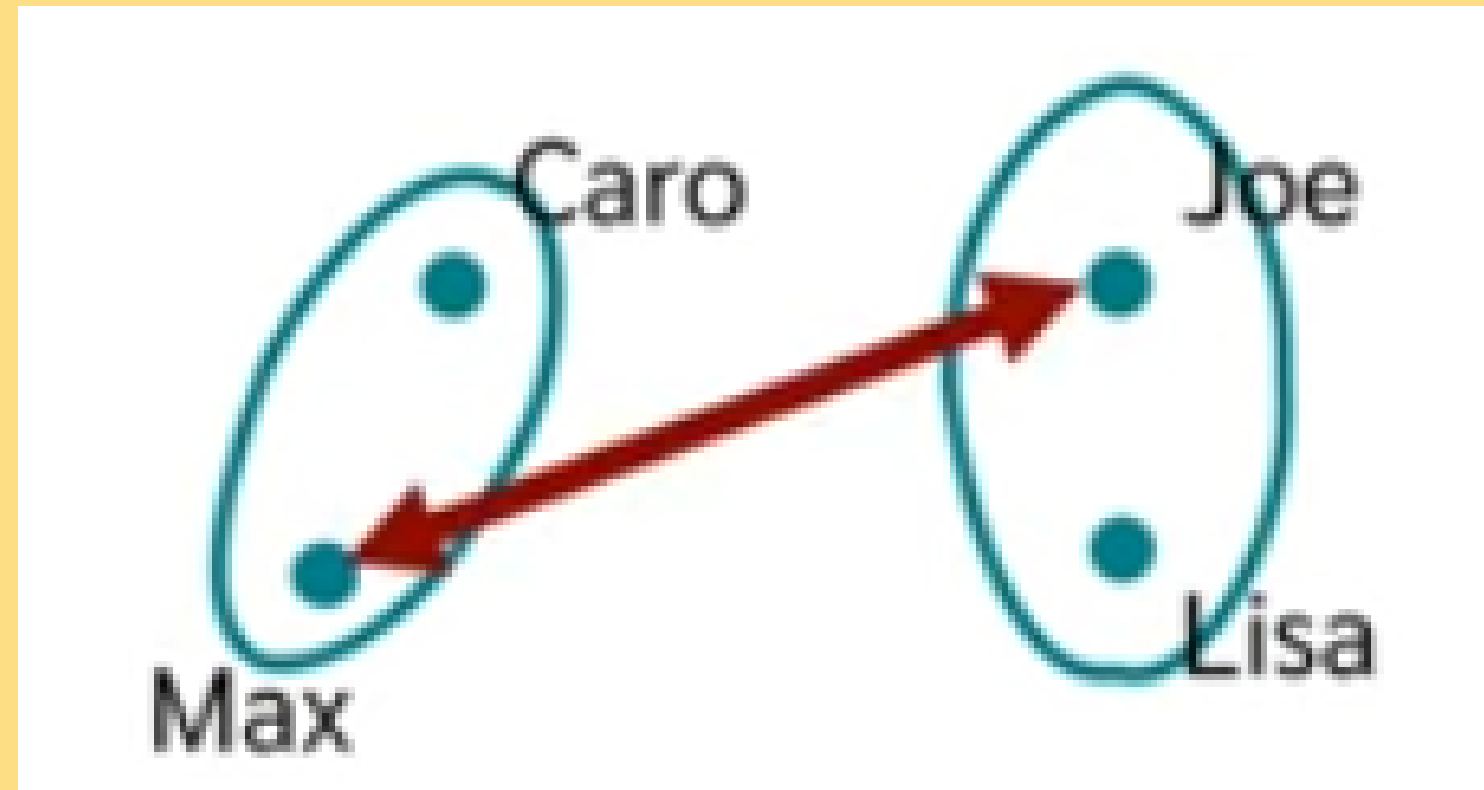
# 1- Single-linkage

- uses the distance between the **closest** elements in the cluster



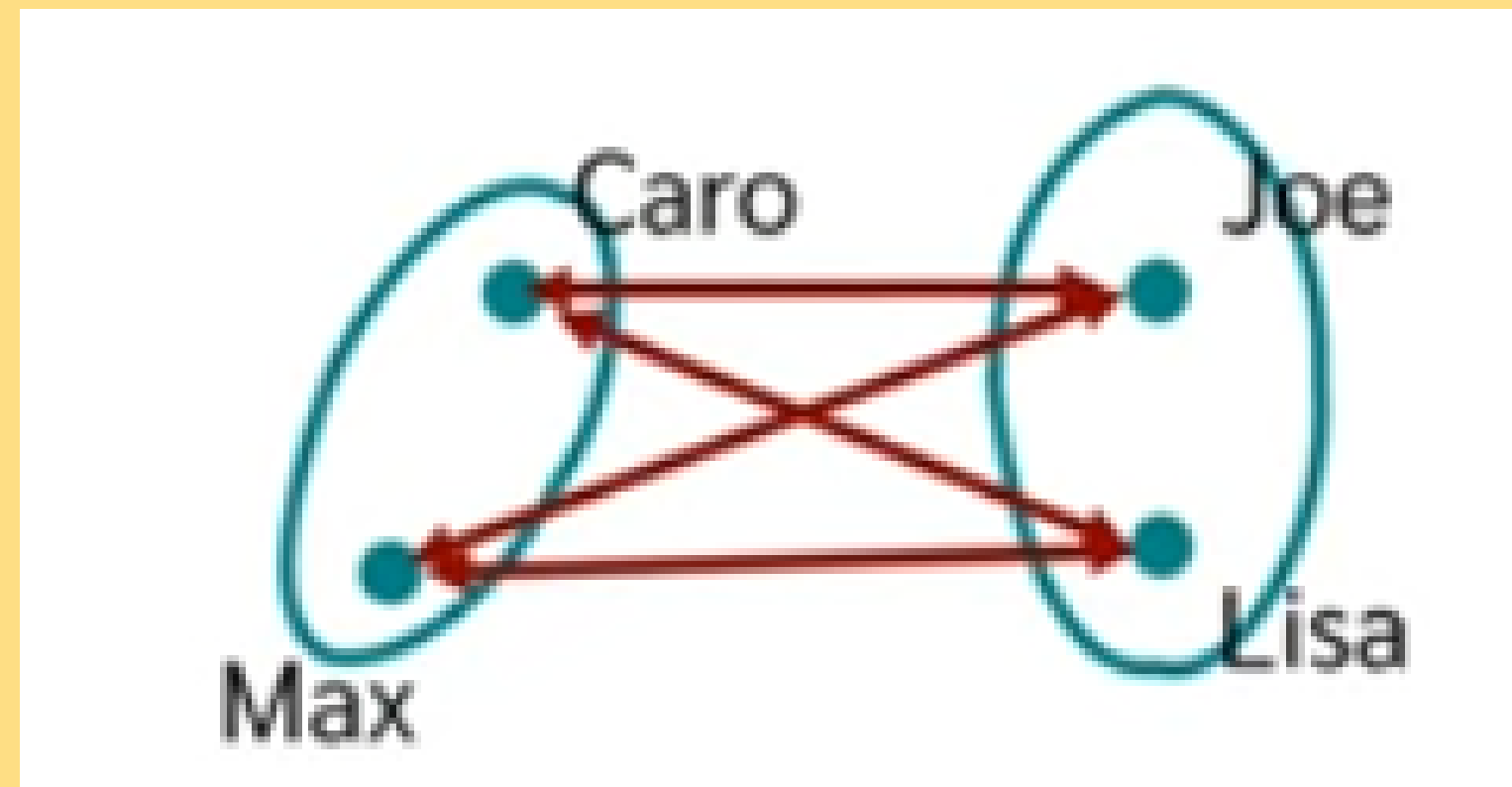
## 2- Complete-linkage

- uses the distance between the **most remote** elements of the cluster.



## 2- Average-linkage

- uses the average of all pairwise distances.

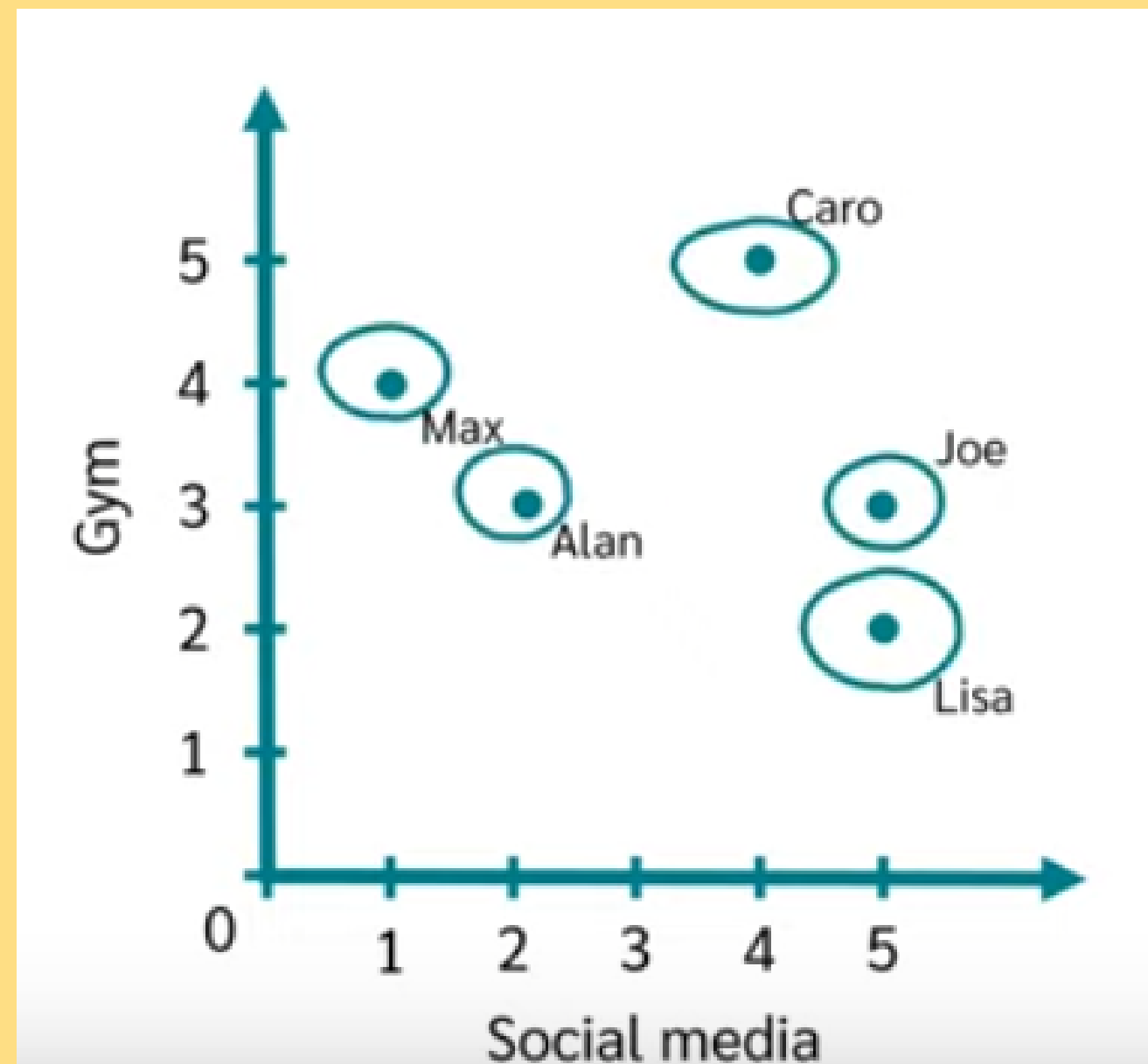


From each combination, the distance is calculated, and then we compute the average of all distances

**Full Example**

# 1- Every single point is a cluster

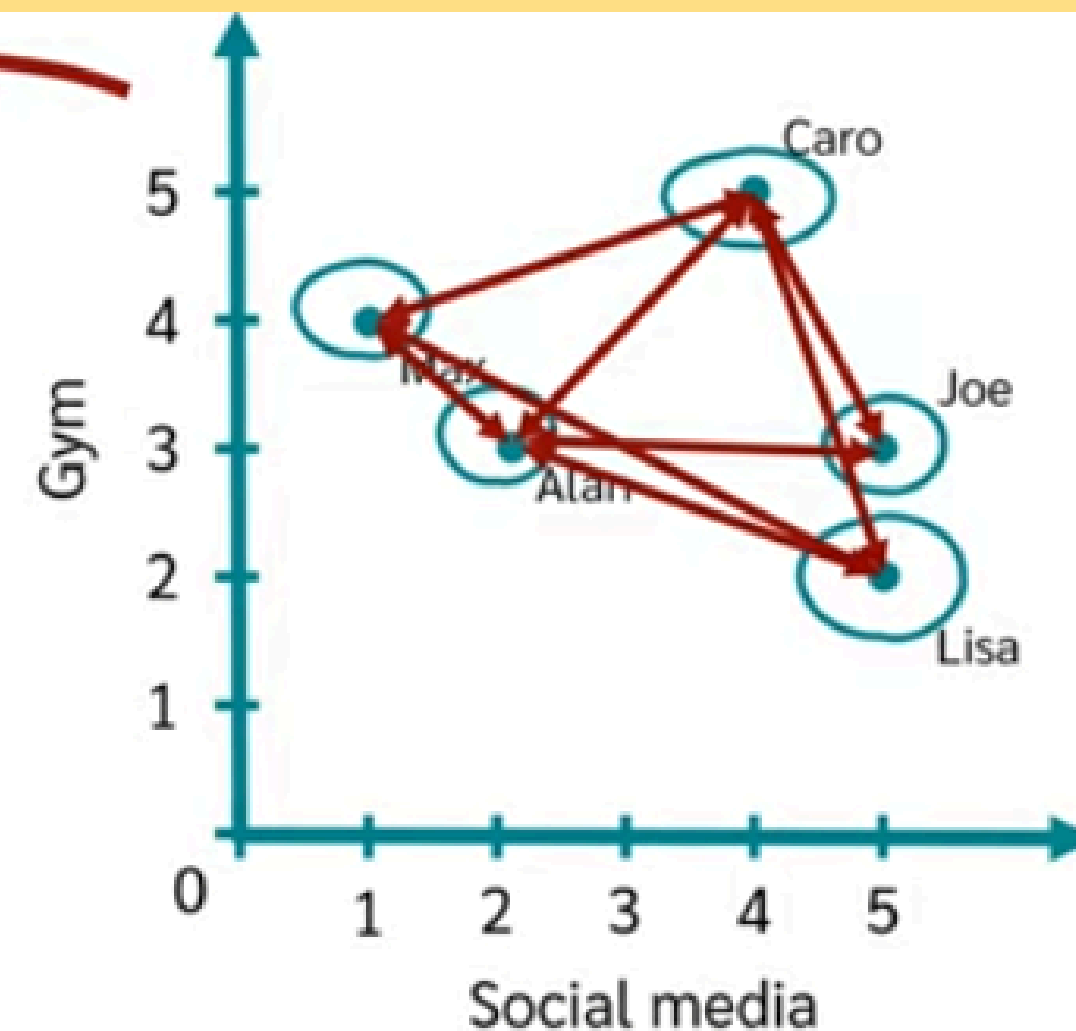
- This is the dataset we are going to work on; we created a cluster for each point.



## 2- Measure the distance between each cluster

- We create a distance matrix that contains the distance between each cluster and the other clusters.

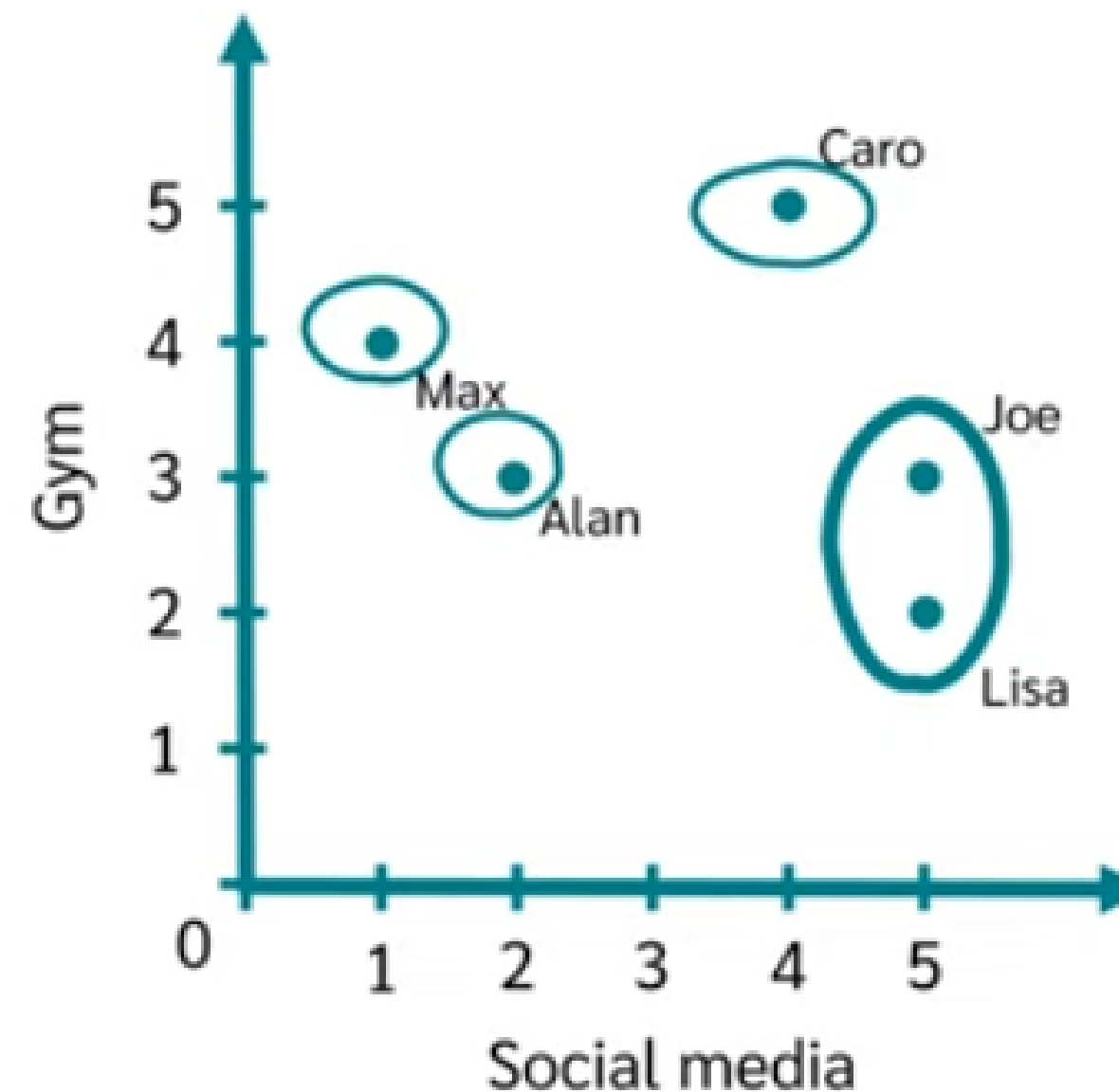
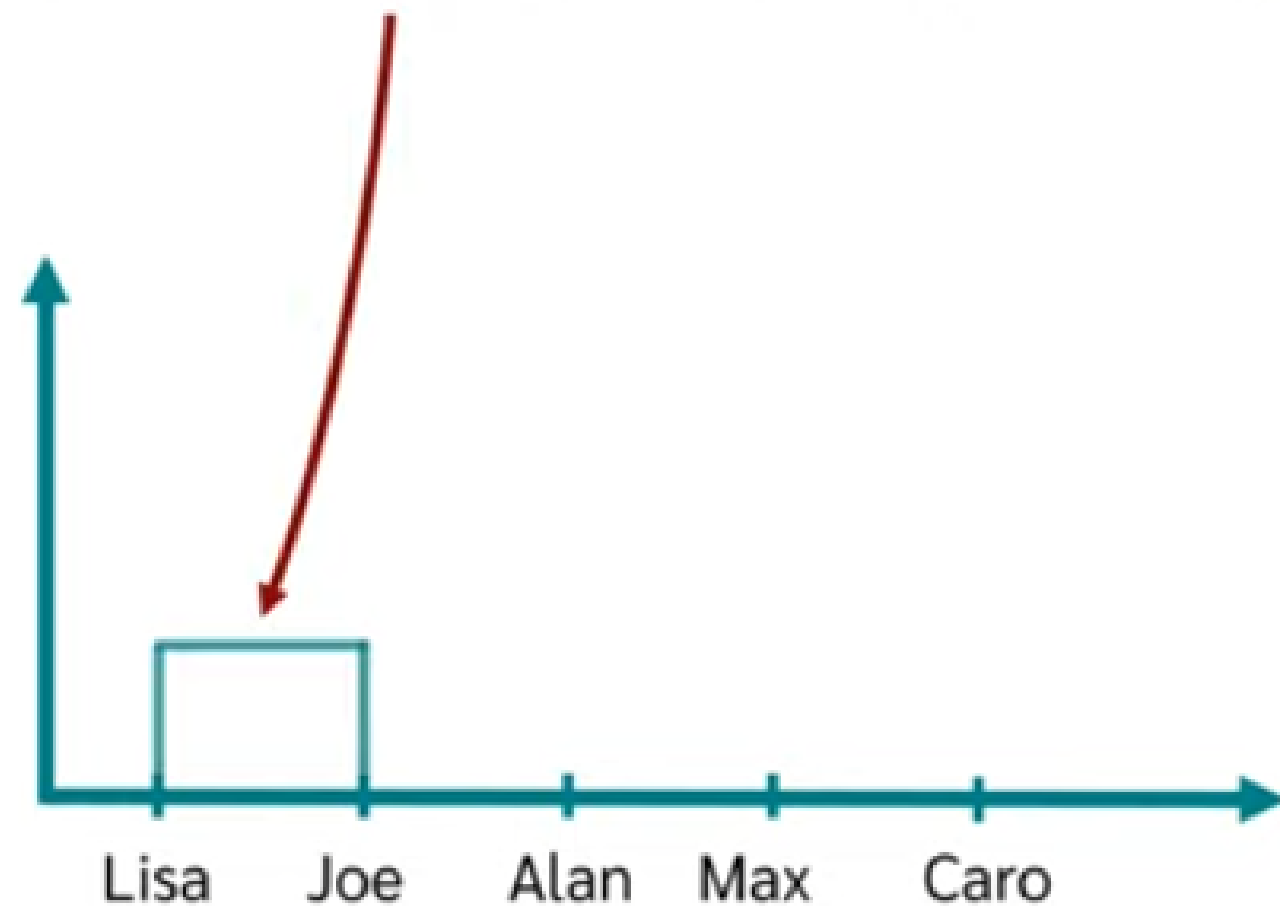
|      | Alan | Lisa | Joe  | Max  | Caro |
|------|------|------|------|------|------|
| Alan | 0    |      |      |      |      |
| Lisa | 3,16 | 0    |      |      |      |
| Joe  | 3,00 | 1,00 | 0    |      |      |
| Max  | 1,41 | 4,47 | 4,12 | 0    |      |
| Caro | 2,83 | 3,16 | 2,24 | 3,16 | 0    |



### 3- Merge the closest clusters

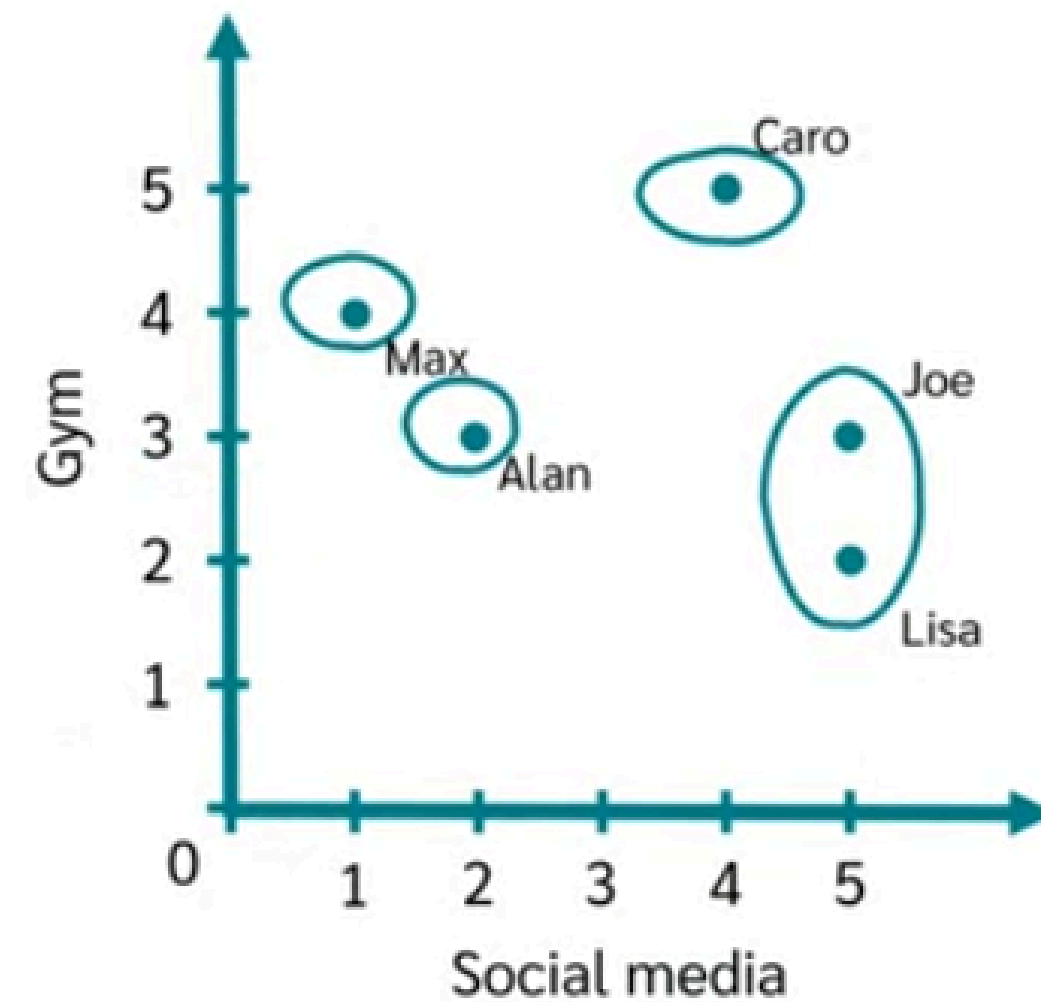
- Lisa and Joe are the closest clusters with a distance of 1.0

In our **tree diagram** or **dendrogram** we can draw the first connection.



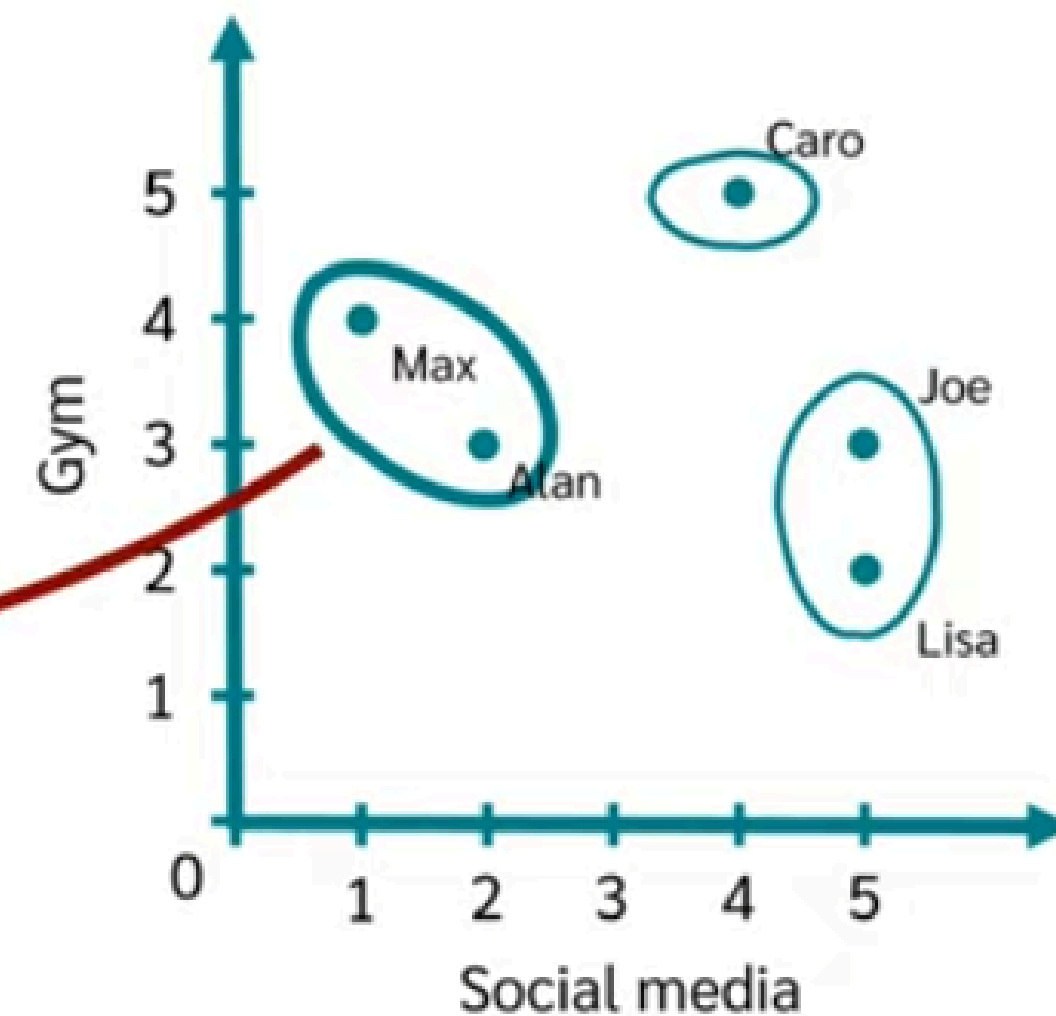
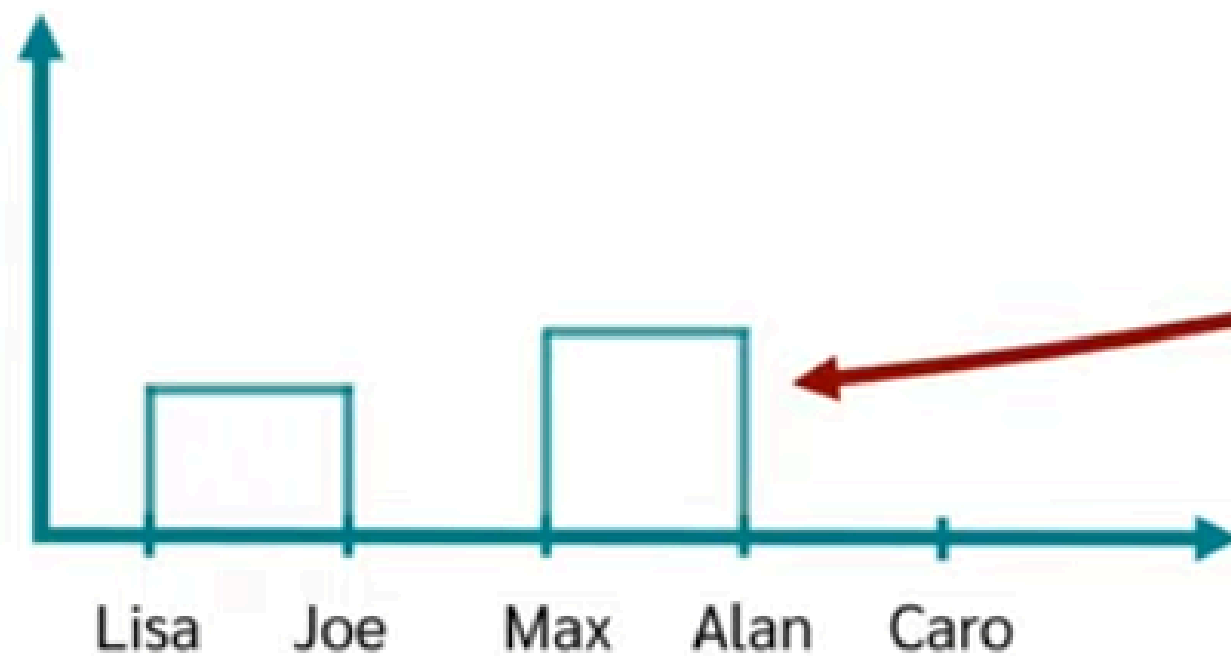
## 4- Update the distance matrix

|           | Alan | Lisa, Joe | Max  | Caro |
|-----------|------|-----------|------|------|
| Alan      | 0    |           |      |      |
| Lisa, Joe | 3,00 | 0         |      |      |
| Max       | 1,41 | 4,12      | 0    |      |
| Caro      | 2,83 | 2,24      | 3,16 | 0    |

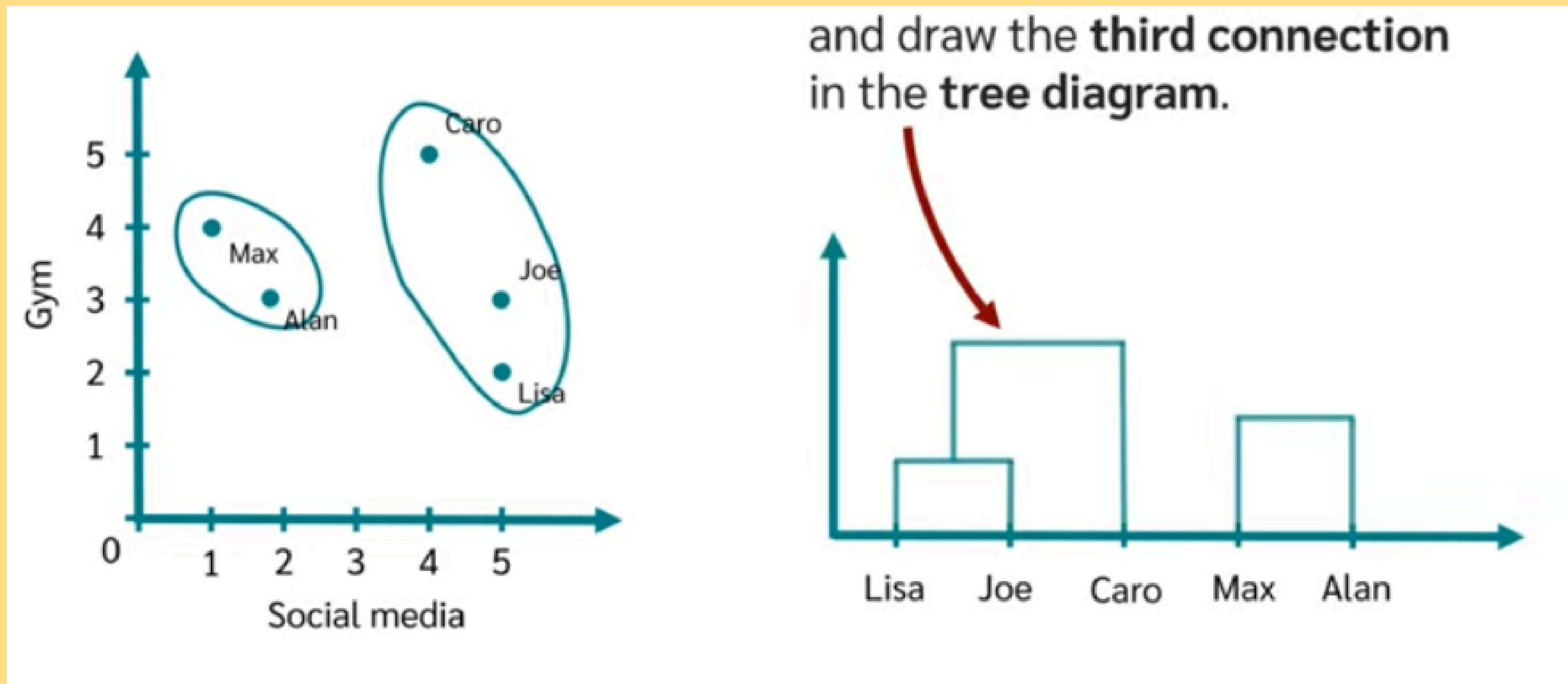


## 5- Choose the new closest clusters

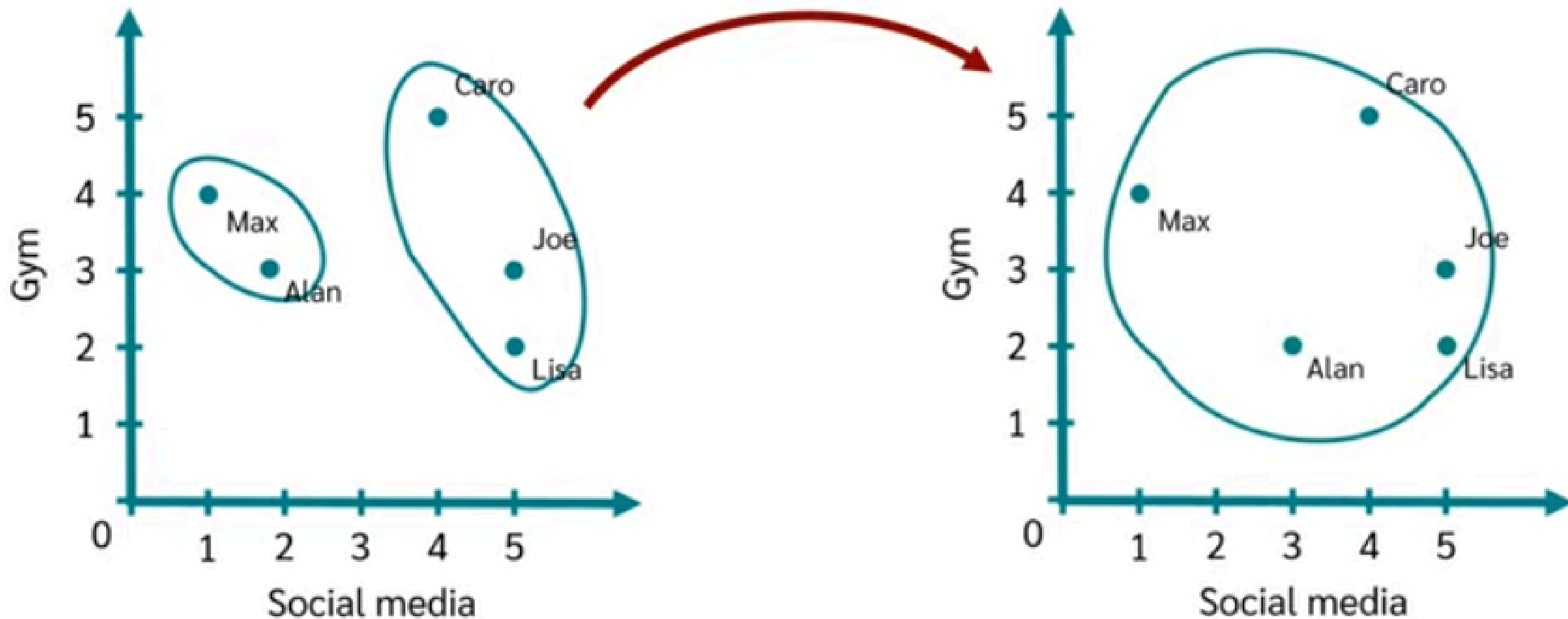
In our **tree diagram** or **dendrogram** we can draw the second connection.



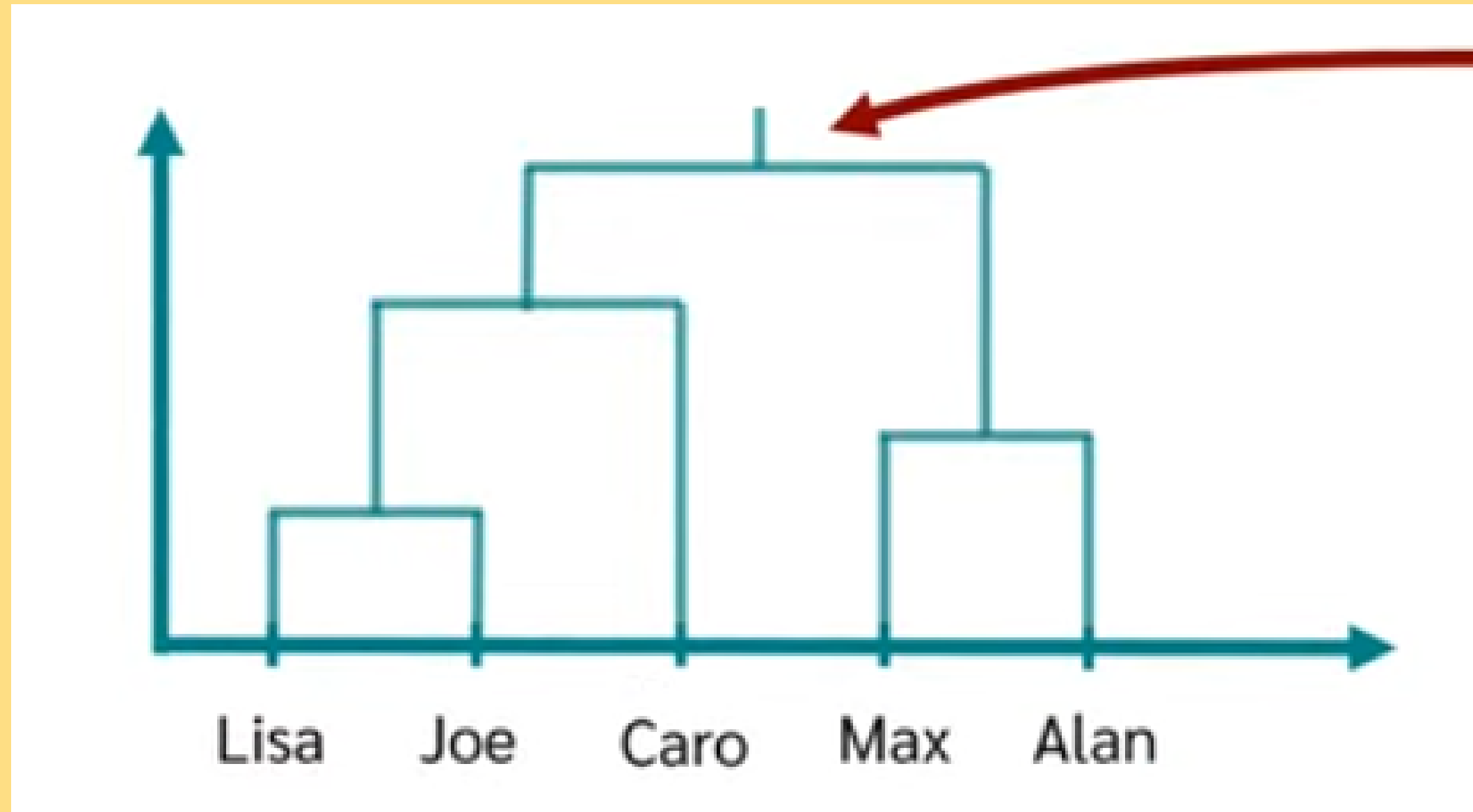
## 6- Continue the same steps until we reach our goal



## 6- Continue the same steps until we reach our goal



## 7- Final Clusters



# K-means vs Hierarchical

| Aspect      | K-means                           | Hierarchical                                                                                                      |
|-------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Complexity  | faster                            | Slower, requires more iterations                                                                                  |
| Scalability | better with large datasets        | Not preferable with large datasets                                                                                |
| Parameters  | Require (K) before clustering     | Does not require parameters                                                                                       |
| Output      | It produces one clustering scheme | It produces a hierarchy of clusters, which can be cut at different levels to obtain different numbers of clusters |

**When** to use which?

# When to use..

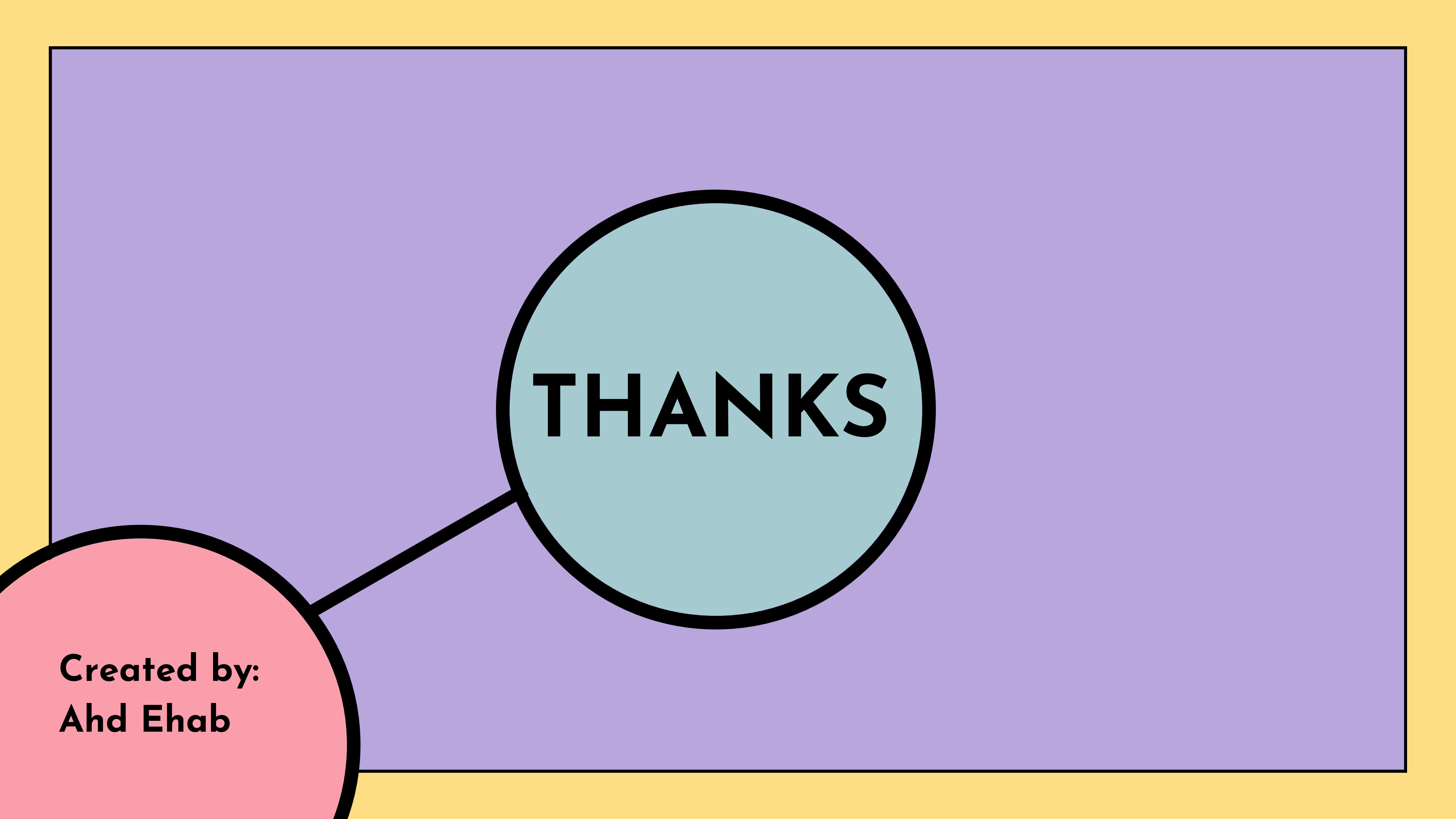
## **K-means:**

- Best for large datasets
- When the number of clusters is known
- When clusters are roughly spherical

## **Hierarchical**

- Good for smaller datasets
- When you want to explore the data to determine the number of clusters
- Better when the dataset is imbalanced

**Any Questions?**



**THANKS**

**Created by:  
Ahd Ehab**